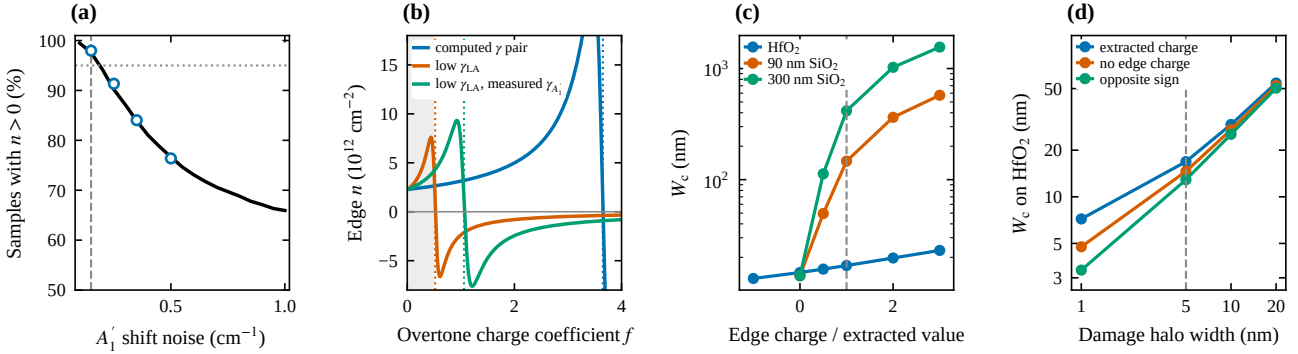
(a)
Samples with $n > 0$ (%)
100
90
80
70
60
50
0.5
1.0
A_1' shift noise (cm^{-1})
(b)
computed γ pair
low γ_{LA}
low γ_{LA}, measured $\gamma_{A_1'}$
Edge n (10^{12} cm^{-2})
15
10
5
0
−5
0
2
4
Overtone charge coefficient f
(c)
HfO$_2$
90 nm SiO$_2$
300 nm SiO$_2$
W_c (nm)
10^3
10^2
0
2
Edge charge / extracted value
(d)
extracted charge
no edge charge
opposite sign
W_c on HfO$_2$ (nm)
50
20
10
5
3
1
5
10
20
Damage halo width (nm)